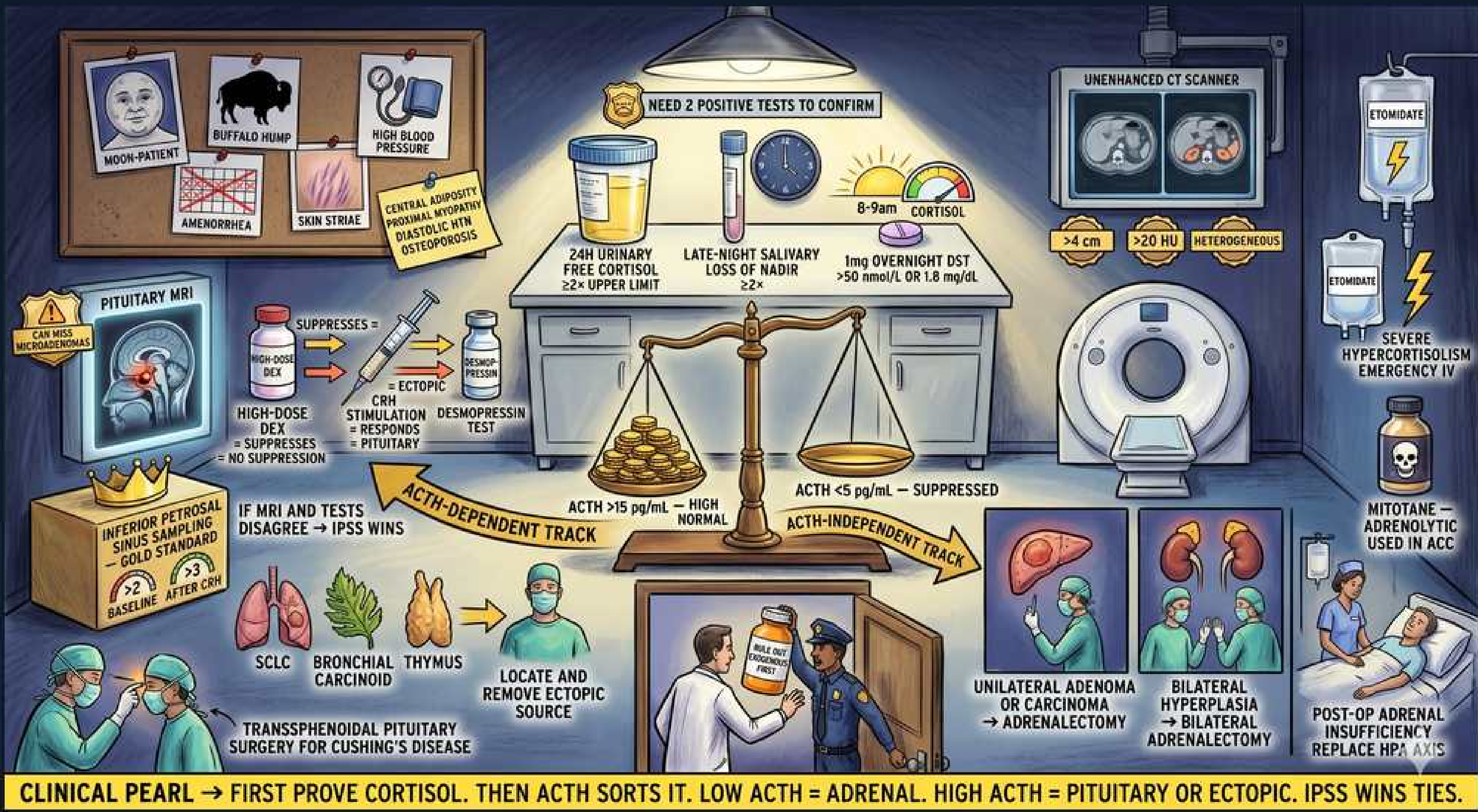
Adrenal — Cushing Diagnostic Lab



1. Need 2 positive tests

WHAT YOU SEE

Banner 'NEED 2 POSITIVE TESTS TO CONFIRM'

WHAT IT ENCODES

STEP 1 of the algorithm: confirm hypercortisolism with at least TWO abnormal FIRST-LINE screening tests before doing anything to localize. The three first-line screens are the 1 mg overnight dexamethasone suppression test, late-night salivary cortisol, and 24-h urine free cortisol. Requiring two reduces false positives from pseudo-Cushing states. Step 0 (implicit) is always excluding EXOGENOUS glucocorticoids by history first.

BOARD TRAP

A single abnormal screen is NOT enough — pseudo-Cushing (alcohol, obesity, depression, illness) causes false positives. Never localize (ACTH, imaging) until cortisol excess is biochemically confirmed on ≥2 tests.

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2. 24h urinary free cortisol

WHAT YOU SEE

Urine jug '24H URINARY FREE CORTISOL ≥2x UPPER LIMIT'

WHAT IT ENCODES

24-hour urine free cortisol measures total unbound cortisol excreted over a day, integrating across cortisol's pulsatile secretion. A value ≥2-4x the upper limit of normal is strongly diagnostic; collect 2-3 separate collections for reliability and check creatinine to confirm complete collection. Useful but can miss mild/cyclic Cushing.

BOARD TRAP

High fluid intake (>5 L/day) and incomplete or over-collection give false results — always verify completeness with urine creatinine. Mild hypercortisolism may give a normal UFC, so a single normal collection doesn't exclude it.

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3. Late-night salivary cortisol

WHAT YOU SEE

Clock/moon 'LATE-NIGHT SALIVARY LOSS OF NADIR'

WHAT IT ENCODES

Late-night (23:00–24:00) salivary cortisol captures the EARLIEST biochemical abnormality in Cushing: loss of the normal circadian NADIR (cortisol should be lowest near midnight). Saliva reflects free, biologically active cortisol; the test is non-invasive and done at home on 2 separate nights — excellent for screening, including cyclic Cushing.

BOARD TRAP

Shift work, irregular sleep, smoking/chewing tobacco, and acute stress disrupt the circadian rhythm and cause false positives — ensure a normal sleep-wake schedule before relying on this test.

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4. 1mg overnight dexamethasone

WHAT YOU SEE

Pill bottle '1mg OVERNIGHT DST >50 nmol/L OR 1.8 ug/dL'

WHAT IT ENCODES

1 mg overnight LOW-DOSE dexamethasone suppression test: give 1 mg dexamethasone at 23:00, measure 08:00 cortisol. A NORMAL axis suppresses to <1.8 µg/dL (<50 nmol/L). FAILURE to suppress (>1.8 µg/dL) is a positive screen. Dexamethasone is used because it suppresses pituitary ACTH without being detected by the cortisol assay.

BOARD TRAP

FALSE POSITIVES from CYP3A4 inducers that speed dex metabolism (phenytoin, rifampin, carbamazepine) and from raised cortisol-binding globulin (estrogen/OCPs, pregnancy). FALSE NEGATIVES with CYP3A4 inhibitors. Stop estrogen ~6 weeks prior or measure free cortisol.

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5. Prove cortisol then ACTH

WHAT YOU SEE

Balance scale with ACTH-dependent vs independent tracks

WHAT IT ENCODES

STEP 2: once cortisol excess is confirmed, measure plasma ACTH (08:00) to split the two tracks — this single test decides the entire downstream workup. ACTH-dependent (high/normal ACTH) → pituitary vs ectopic. ACTH-independent (suppressed ACTH) → adrenal. The scale image captures the whole 'first prove cortisol, then ACTH sorts it' logic.

BOARD TRAP

ACTH is labile (degrades quickly) — must be collected cold and processed promptly or you get a falsely low value that mislabels a case as adrenal. An inappropriately NORMAL ACTH with proven cortisol excess counts as ACTH-DEPENDENT.

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6. ACTH high/normal → dependent

WHAT YOU SEE

Left track 'ACTH >15 pg/mL HIGH NORMAL'

WHAT IT ENCODES

ACTH not suppressed (typically >15-20 pg/mL, or inappropriately normal) → ACTH-DEPENDENT Cushing → source is PITUITARY (Cushing disease, ~80%) or ECTOPIC (~20%). Next steps: pituitary MRI plus dynamic testing (high-dose dexamethasone suppression and CRH stimulation), and IPSS if those are discordant.

BOARD TRAP

An 'inappropriately normal' ACTH in the face of high cortisol is abnormal and counts as ACTH-DEPENDENT — autonomous adrenal disease would have fully SUPPRESSED ACTH. Don't read a normal-range ACTH as reassuring here.

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7. ACTH low → independent

WHAT YOU SEE

Right track 'ACTH <5 pg/mL SUPPRESSED'

WHAT IT ENCODES

ACTH suppressed (typically <5-10 pg/mL) → ACTH-INDEPENDENT Cushing → ADRENAL source making cortisol autonomously → go directly to ADRENAL CT. The non-tumor adrenal tissue is atrophic from chronic ACTH suppression, so plan perioperative glucocorticoid cover and a taper after adrenalectomy.

BOARD TRAP

Low ACTH means SKIP the pituitary/dynamic tests entirely and image the adrenals — ordering high-dose dex or CRH in a suppressed-ACTH patient is a wasted, wrong step.

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8. High-dose dex / CRH

WHAT YOU SEE

'HIGH-DOSE DEX SUPPRESSES', 'CRH STIMULATION RESPONDS = PITUITARY', desmopressin test

WHAT IT ENCODES

Dynamic tests to separate PITUITARY from ECTOPIC within the ACTH-dependent track: a pituitary adenoma retains partial feedback — it SUPPRESSES cortisol >50% on HIGH-dose (8 mg) dexamethasone and RESPONDS (ACTH/cortisol rise) to CRH (or desmopressin). Ectopic tumors typically do NEITHER (autonomous).

BOARD TRAP

Bronchial CARCINOID is the classic exception — it can suppress with high-dose dex and respond to CRH, mimicking pituitary disease and fooling these tests. That discordance is exactly why IPSS is the gold standard tie-breaker.

9. Pituitary MRI

WHAT YOU SEE

Brain MRI image labeled 'PITUITARY MRI'

WHAT IT ENCODES

In ACTH-dependent Cushing, pituitary MRI (with dynamic gadolinium) looks for a corticotroph adenoma. A definite adenoma >6 mm with concordant dynamic tests confirms Cushing disease and directs transsphenoidal surgery.

BOARD TRAP

Most corticotroph adenomas are MICROadenomas (<6 mm) and up to ~40% are NOT visible on MRI — a normal MRI does NOT exclude Cushing disease. Conversely, incidental non-functioning pituitary lesions are common, so a small MRI 'lesion' isn't automatically the culprit; confirm with IPSS.

10. IPSS gold standard

WHAT YOU SEE

'INFERIOR PETROSAL SINUS SAMPLING GOLD STANDARD'

WHAT IT ENCODES

Inferior petrosal sinus sampling (IPSS) is the GOLD STANDARD for distinguishing pituitary from ectopic ACTH when MRI and dynamic tests are ambiguous. Catheters sample ACTH from both petrosal sinuses and a peripheral vein, before and after CRH. A central:peripheral ACTH gradient (≥2 basal, ≥3 post-CRH) = PITUITARY source; no gradient = ECTOPIC.

BOARD TRAP

If MRI shows a small lesion but the biochemistry is discordant, IPSS overrides imaging — IPSS WINS TIES. A 'lesion' on MRI without a central gradient on IPSS is likely an incidentaloma, not the source.

11. Ectopic source hunt

WHAT YOU SEE

'SCLC / BRONCHIAL / THYMUS', 'LOCATE AND REMOVE ECTOPIC SOURCE'

WHAT IT ENCODES

No central IPSS gradient → ECTOPIC ACTH → hunt the tumor: small-cell lung carcinoma (most common), bronchial/thymic carcinoid, medullary thyroid carcinoma, pheochromocytoma, pancreatic NET. Use chest/abdomen CT, and functional imaging (Ga-68 DOTATATE for neuroendocrine tumors, FDG-PET). Definitive treatment is locating and resecting the source.

BOARD TRAP

If the source can't be found or surgery isn't possible, control hypercortisolism medically (ketoconazole, metyrapone, osilodrostat, etomidate IV for crisis) or by bilateral adrenalectomy — don't leave severe hypercortisolism untreated while hunting the tumor.

12. Adrenal CT for independent

WHAT YOU SEE

Unenhanced CT scanner, '>4cm', '>20 HU', 'HETEROGENEOUS'

WHAT IT ENCODES

ACTH-INDEPENDENT (suppressed ACTH) → adrenal CT (unenhanced + washout protocol). Findings: a unilateral homogeneous lipid-rich (<10 HU, <4 cm) lesion = benign adenoma → unilateral adrenalectomy; a large (>4 cm), heterogeneous, lipid-poor (>20 HU), slow-washout mass = adrenocortical CARCINOMA; bilateral nodules = macronodular hyperplasia or PPNAD.

BOARD TRAP

After unilateral adrenalectomy the suppressed contralateral gland is atrophic → expect POST-OP ADRENAL INSUFFICIENCY; give perioperative steroids and taper slowly. Don't forget glucocorticoid replacement.

13. Etomidate IV (crisis control)

WHAT YOU SEE

IV bag 'ETOMIDATE', 'SEVERE HYPERCORTISOLISM EMERGENCY IV'

WHAT IT ENCODES

For SEVERE acute hypercortisolism (e.g., psychosis, severe hypokalemia/HTN, sepsis from ectopic ACTH), IV ETOMIDATE rapidly blocks adrenal 11β-hydroxylase and lowers cortisol within hours — the fastest medical control while definitive workup/surgery is arranged. Oral steroidogenesis inhibitors (ketoconazole, metyrapone, osilodrostat) are used for ongoing control.

BOARD TRAP

Etomidate suppresses cortisol so effectively it can precipitate ADRENAL INSUFFICIENCY — it must be titrated with cortisol monitoring and a 'block-and-replace' strategy, not given blindly.

14. Mitotane (adrenolytic in ACC)

WHAT YOU SEE

Skull bottle 'MITOTANE ADRENOLYTIC IN ACC', 'POST-OP ADRENAL INSUFFICIENCY REPLACE'

WHAT IT ENCODES

MITOTANE is an adrenolytic (adrenal-cytotoxic) drug used as adjuvant therapy for adrenocortical CARCINOMA after resection and for unresectable/metastatic disease (often with etoposide-doxorubicin-cisplatin). It destroys adrenocortical cells, so it causes adrenal insufficiency by design and requires glucocorticoid (and often mineralocorticoid) replacement.

BOARD TRAP

Mitotane induces hepatic CYP3A4 and raises cortisol-binding globulin, so patients need HIGHER-than-usual hydrocortisone replacement doses — under-replacing precipitates adrenal crisis. It is specific to ACC, not a treatment for benign adenoma or pituitary disease.

15. Clinical pearl

WHAT YOU SEE

'FIRST PROVE CORTISOL. THEN ACTH SORTS IT. LOW=ADRENAL, HIGH=PITUITARY/ECTOPIC. IPSS WINS TIES'

WHAT IT ENCODES

The locked diagnostic sequence: (0) exclude exogenous steroids → (1) PROVE cortisol excess with ≥2 first-line screens → (2) measure ACTH to sort tracks → LOW ACTH = ADRENAL (adrenal CT), HIGH/normal ACTH = PITUITARY or ECTOPIC (MRI + high-dose dex/CRH) → (3) IPSS WINS TIES when MRI and dynamic tests disagree, distinguishing pituitary (central gradient) from ectopic.

BOARD TRAP

The recurring board traps: imaging BEFORE biochemical confirmation, checking ACTH before proving cortisol, trusting a normal pituitary MRI to exclude Cushing disease, and being fooled by a bronchial carcinoid that mimics pituitary on dynamic tests — IPSS settles that last one.